

ALEX MORGAN

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Computer Science; Minor in Mathematical Sciences

Expected May 2027

- GPA:** 3.82 / 4.00 | **Honors:** Dean's List (All Semesters)
- Relevant Coursework:** Distributed Systems, Algorithm Design & Analysis, Operating Systems, Principles of Functional Programming, Machine Learning, Database Systems, Object-Oriented Software Engineering.

TECHNICAL SKILLS

- Languages:** Python, Java, C, C++, TypeScript, JavaScript, SQL (PostgreSQL, MySQL), HTML/CSS, Go
- Frameworks:** React.js, Node.js (Express), FastAPI, Flask, Spring Boot, JUnit, Next.js
- Tools & Infra:** Docker, Git, AWS (S3, EC2, Lambda), PostgreSQL, Redis, MongoDB, Linux/Unix, CI/CD (GitHub Actions)
- Methodologies:** Agile/Scrum, RESTful API Design, Object-Oriented Programming (OOP), Unit & Integration Testing

PROFESSIONAL EXPERIENCE

Software Engineering Intern

May 2025 – August 2025

Nexus Cloud Technologies

Seattle, WA (Remote)

- Engineered and deployed a microservice architecture using **FastAPI** and **Docker** to aggregate real-time system metrics, processing over 1.2M daily log events and lowering data visualization latency by 34%.
- Optimized slow database workflows by redesigning relational schemas and adding targeted indexing in **PostgreSQL**, cutting API endpoint turnaround response times from 420ms down to 85ms.
- Built an end-to-end automated testing matrix using **PyTest** inside a **GitHub Actions** CI/CD pipeline, raising overall test coverage from 68% to 91% and saving engineering teams ~5 deployment hours weekly.
- Collaborated actively inside a 6-person Agile sprint squad, contributing to daily standups, code reviews, and cross-functional product alignments.

Undergraduate Teaching Assistant (Computer Science Department)

August 2025 – Present

Carnegie Mellon University

Pittsburgh, PA

- Deliver weekly recitations and office hours clarifying foundational data structures, algorithmic complexity, and dynamic programming paradigms for a cohort of 60+ undergraduates.
- Evaluate and provide detailed design feedback on weekly code submissions written in **C** and **Python**, debugging intricate memory management and pointer logic vulnerabilities.

ENGINEERING PROJECTS

Automated Resume Parser & Semantic Ranking Engine

March 2026 – April 2026

Personal Project

[GitHub Link](#)

- Architected a full-stack asynchronous parsing tool leveraging **React**, **TypeScript**, **FastAPI**, and a **PostgreSQL** cluster.
- Integrated **Sentence-BERT** text embeddings via Hugging Face to compare unstructured resume PDF data against explicit job description vectors, yielding semantic alignment scores.
- Leveraged **Redis** task queues to gracefully handle multi-file uploads concurrently, executing batch background extraction requests smoothly without thread blocking.

OpenCV Smart Recycling Sorter & Hardware Router

January 2026

Build18 Hackathon (Award: Best Automation System)

[GitHub Link](#)

- Developed a localized computer-vision sorting pipeline using **Python** and **OpenCV** to categorize trash objects (paper, plastic, metal) arriving on a physical conveyor mechanism.
- Trained a custom, lightweight Convolutional Neural Network (CNN) reaching 94.2% classification accuracy on edge hardware, streaming real-time control directives to an Arduino micro-controller.
- Programmed the responsive micro-controller logic in **C++** to activate a passive physical diverter slide, cleanly filtering incoming items into distinct bins within 250ms of visual capture.

LEADERSHIP & ACTIVITIES

Executive Organizing Officer

September 2024 – Present

Build18 Hardware & ECE Hackathon

Pittsburgh, PA

- Co-manage logistics, budgeting, and corporate sponsorships for CMU's premier annual hardware hackathon, serving over 300+ cross-departmental student participants.
- Maintained and updated the core organizational web registry platform using **Next.js** and **Node.js**, streamlining applicant registrations.